Tangent Approximation based Variational InferencE for Strongly Super-Gaussian likelihoods

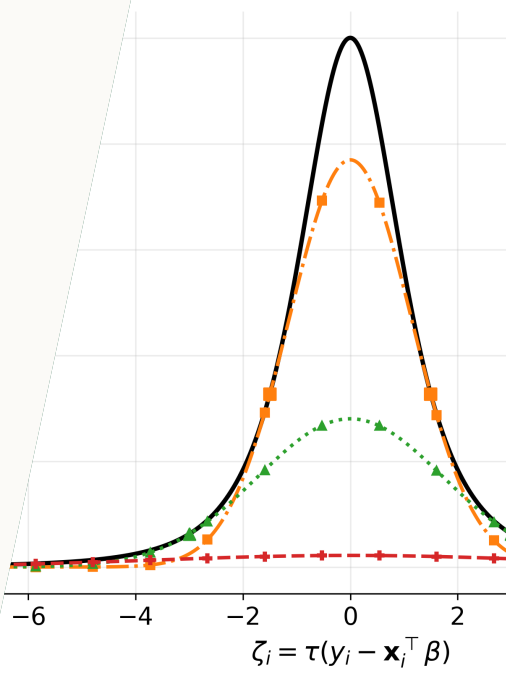
Scalable approximate Bayesian computation
with convergence and variational-risk guarantees

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1
Background and Motivation

1 A Large Sample Size Problem

Bayesian quantile regression for U.S. Census 2000 data

5% sample of U.S. population, contains **salary** and **demographic data**

Inferential goal: Infer effects of demographic factors across the distribution of salaries.

$$Q_{\log(\text{salary})}(u \mid \mathbf{x}_i) = \mathbf{x}_i^{\top} \beta_u, \qquad u \in \{0.05, 0.10, \ldots, 0.95\}.$$

5×10^6
Census records

11
encoded predictors

19
quantile levels

19 posterior inference problems, each involving five million observations.

Posterior sampling algorithms start struggling at this scale.

U.S. Census 2000 5% sample; Yang, Meng, and Mahoney (2013).

1 Computation vs. Uncertainty: Why not both?

Monte Carlo Methods

Monte Carlo based sampling to explore the posterior distribution

- Posterior samples
- Rich uncertainty quantification
- ✕ Prohibitive for large sample sizes

MAP Estimation

Direct optimization of (unnormalized) posterior distribution

- A single posterior mode
- Very fast optimization
- ✕ No uncertainty quantification by itself

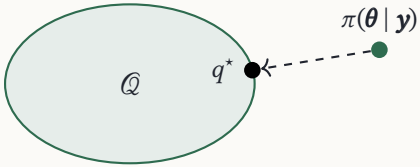
Can optimization retain a distributional approximation?

Variational Inference (VI)

Approximate the posterior distribution through optimization over a family of specified distributions $\mathcal{Q}$.

$$q^\star(\boldsymbol{\theta}) = \arg\min_{q \in \mathcal{Q}} \mathrm{D}\left[q(\boldsymbol{\theta}) \,\|\, \pi(\theta \mid \boldsymbol{y})\right]$$

For instance, D may be the Kullback-Liebler (KL) divergence.



Scalability of optimization—without reducing inference to a point.

1 Existing VI Methods: Different Trade-offs

A VI method involves two distinct choices: **the approximation** and **how it is optimized**.

Literature has many important classes of VI (Blei *et al.*, 2017), but two of the most popular are:

Mean-Field VI

Parisi (1998); Bhattacharya *et al.* (2023)

$$\mathcal{Q}_{\text{MF}} = \left\{ q(\boldsymbol{\theta}) = \prod_{j=1}^{d} q_j(\theta_j) \right\}$$

- Scalable coordinate-ascent updates
- Well-studied convergence properties
- ✗ Posterior dependence is removed
- ✗ Limited applicability

Black-Box VI

Kucukelbir *et al.* (2017).

$$\nabla_{\boldsymbol{\theta}} \mathscr{L}(\boldsymbol{\theta}) \approx \left\{ \frac{1}{S} \sum_{s=1}^{S} \widehat{\nabla}_{\boldsymbol{\theta}} \mathscr{L}_s(\boldsymbol{\theta}) \right\}$$

- Broadly applicable; mean-field or richer families
- Uses automatic differentiation for optimization
- ✗ Expensive, noisy Monte-Carlo based gradients
- ✗ Lacks convergence guarantees

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x VI Kucukelbir *et al.* (2017).

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Question

Can likelihood structure be exploited for deterministic, scalable VI, with convergence guarantees without restrictive mean-field assumptions?

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Literature has many important [illegible] but two of the most popular are:

Mean-

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- × P
- × Limited applicability

Tangent Approximation

Main Idea: Parametrized tangent lower bounds of the log-likelihood, followed by optimization over said parameters.

- Conceptualized for **logistic regression** by Jaakkola and Jordan (2000).
- Guarantees for **integrated variational risk** and **local algorithmic stability** established (Ghosh *et al.*, 2022).
- Has been largely restricted to logistic regression and related models.

r et al. (2017).

amilies

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× Lacks convergence guarantees

2
Tangent Approximation based VI

2 Strongly Super-Gaussian Likelihoods

Consider a dataset consisting of n units with response vector $\mathbf{y} = (y_1, \ldots, y_n)^\top$ and a corresponding design matrix $\mathbf{X} = (\mathbf{x}_1, \ldots, \mathbf{x}_n)^\top$.

Definition

The conditional likelihood of y_i given $\mathbf{x}_i$ is strongly super-Gaussian (SSG) if

$$\log p(y_i \mid \mathbf{x}_i, \boldsymbol{\theta}) = s_i \zeta_i + t_i h(\zeta_i^2),$$

where ζ_i is $\mathbf{x}_i^\top \beta$ or an affine transform and h convex, decreasing and twice-differentiable on $\mathbb{R}^+$.

We exploit **Strong Super-Gaussianity** of likelihoods for **tangent approximation**.

Consid... ...nding design...

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Tangent lower bound

For any $\xi_i > 0$, by the convexity of h, we have

$$h(\zeta_i^2) \geq \underbrace{h'(\xi_i^2)}_{\text{negative}} \zeta_i^2 + \underbrace{h'(\xi_i^2) - h'(\xi_i^2)\xi_i^2}_{\gamma(\xi_i)}.$$

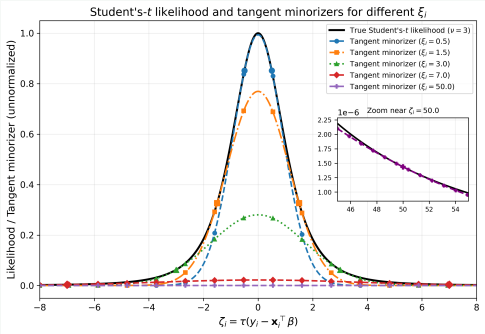
- These lower bounds form a family of downward facing parabolas, which are tangent to $h(\zeta_i^2)$ at $|\zeta_i| = \xi_i$.
- Replacing $h(\zeta_i^2)$ by the lower bound in the **SSG** likelihood $p(y_i \mid \mathbf{x}_i, \boldsymbol{\theta})$ yields a tangent minorizer $\varphi(y_i \mid \mathbf{x}_i, \boldsymbol{\theta}, \xi_i)$.

2 Two Classes of SSG Likelihoods

Despite the general formulation, we focus on two concrete **SSG** likelihood classes.

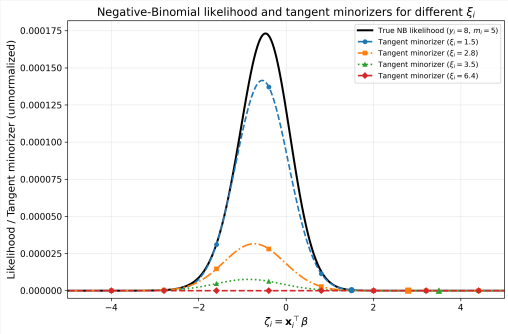
Type I: Heavy-Tailed Regression

Examples: Laplace and Student's-*t* regression, Gaussian scale-mixture errors, quantile regression.



Type II: Bernoulli-Type Regression

Examples: Logistic, Binomial and Negative-Binomial regression.



2 Fractional likelihood and working likelihood

For fixed $\alpha \in (0, 1]$, we temper the likelihood and then apply tangent lower bounds:

$$\underbrace{p_\alpha(\mathbf{y} \mid \mathbf{X}, \boldsymbol{\theta}) = \left\{ \prod_{i=1}^{n} p(y_i \mid \mathbf{x}_i, \boldsymbol{\theta}) \right\}^\alpha}_{\text{true fractional likelihood}} \quad \Longrightarrow \quad \underbrace{\varphi_\alpha(\mathbf{y} \mid \mathbf{X}, \boldsymbol{\theta}, \boldsymbol{\xi}) = \left\{ \prod_{i=1}^{n} \varphi(y_i \mid \mathbf{x}_i, \boldsymbol{\theta}, \xi_i) \right\}^\alpha}_{\text{tangent working likelihood}},$$

where $\boldsymbol{\xi} = (\xi_1, \ldots, \xi_n)^\mathsf{T}$.

$\alpha = 1$: **standard likelihood** · $0 < \alpha < 1$: **tempered likelihood**

By this construction, $\varphi_\alpha(y \mid \mathbf{X}, \boldsymbol{\theta}, \boldsymbol{\xi})$

- is a lower bound of $p_\alpha(y \mid \mathbf{X}, \boldsymbol{\theta})$.
- has exponential negative-quadratic form like a Gaussian kernel.

The working likelihood is conjugate to Gaussian priors!

2 Variational posterior distribution

Combining α-fractional tangent working likelihood with **conjugate Gaussian prior** $\pi(\boldsymbol{\theta})$ yields:

Closed form α-fractional variational posterior distribution

$$\pi_\alpha(\boldsymbol{\theta} \mid \mathscr{D}_n, \boldsymbol{\xi}) = \frac{\varphi_\alpha(\mathbf{y} \mid \mathbf{X}, \boldsymbol{\theta}, \boldsymbol{\xi})\, \pi(\boldsymbol{\theta})}{\exp\{\mathrm{L}(\boldsymbol{\xi})\}},$$

where $\mathrm{L}(\boldsymbol{\xi}) = \log \int_\Theta \varphi_\alpha(\mathbf{y} \mid \mathbf{X}, \boldsymbol{\theta}, \boldsymbol{\xi})\, \pi(\boldsymbol{\theta})\, d\boldsymbol{\theta}$ is the Evidence Lower Bound (**ELBO**).

Optimal choice of the variational parameter vector $\boldsymbol{\xi} \Longleftarrow$ maximizing the ELBO $\mathrm{L}(\boldsymbol{\xi})$.

2 Variational posterior distribution

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Optimization of ξ

We use a **variational Expectation-Maximization (EM)** algorithm.

- Alternates between **closed-form updates** of the variational posterior and tangent parameters.
- Variational parameter **updates are independent for each data point**, making the algorithm scalable in n, and parallelizable at each iteration.
- Total computation **cost per iteration**: $\mathcal{O}(np^2 + p^3)$.
- **Entirely deterministic**: no Monte Carlo gradients, learning rates, or automatic differentiation.

We refer to this general methodology as **TAVIE-SSG**.

3

Theoretical Guarantees

3 Convergence of the TAVIE-SSG Algorithm

Theorem (Convergence of the TAVIE-SSG EM algorithm)

Under mild regularity assumptions, for both Type I and Type II SSG likelihoods, the sequence of TAVIE-SSG iterates converges to a fixed point $\boldsymbol{\xi}^\star \in \mathbb{R}_+^n$. *Moreover, every such fixed point is a critical point of the profile ELBO:*

$$\nabla_{\boldsymbol{\xi}} \mathsf{L}(\boldsymbol{\xi}^\star) = \mathbf{0}.$$

- The guarantee holds from any initialization and applies to both classes of SSG likelihoods.
- Convergence is established despite the generally nonconvex profile ELBO.
- The limiting point is stationary, but need not be the unique or global maximizer.

3 Variational Risk Bounds

Theorem (Variational risk bounds for Type I and Type II SSG likelihoods)

Under mild regularity conditions, for any $0 < \varepsilon_n < 1/3$, the variational risk satisfies with $\mathbb{P}_{\theta_0}$-probability at least $(1 - 2\varepsilon_n - [(D-1)^2 n\varepsilon_n^2]^{-1}$:

$$\int_{\beta \in \Theta} D_\alpha(\beta \parallel \beta_0)\pi_\alpha(\boldsymbol{\theta} \mid \mathscr{D}_n, \xi^\star)d\beta \leq D\varepsilon_n^2 + \frac{p \log p}{n} + \frac{Cp}{n} \log\left(\frac{1}{\varepsilon_n}\right),$$

for arbitrary $D > 1$, $c \in \mathbb{R}^+$, and $C \in \mathbb{R}^+$ depending only on prior hyperparameters, design matrix, and true model parameters.

- $0 < \alpha < 1$: scaled α-Rényi divergence, $D_\alpha = (1-\alpha)D_\alpha$.
- $\alpha = 1$: scaled squared Hellinger distance, $D_1 = cH^2$.
- These risk bounds are non-asymptotic in all cases and taking $\varepsilon_n^2 \asymp n^{-1} p \log n$, the resulting risk bound is $\mathcal{O}(n^{-1}p)$, up to logarithmic factors, yielding near-minimax optimality.

4
Empirical Evaluation

4 Competing Methods and Evaluation Criteria

Competing Methods

1. Automatic Differentiation VI (ADVI) – Mean-Field (Kucukelbir *et al.*, 2017)
2. ADVI – Full-Rank (FR) (Kucukelbir *et al.*, 2017)
3. Deterministic ADVI (DADVI) (Giordano *et al.*, 2024)
4. Mean-Field VI (MFVI) whenever available
5. PyMC No-U-Turn Sampler (NUTS)

Evaluation Criteria

- **Point-estimation accuracy**: Mean Squared Error (MSE)
- **Posterior Fidelity**: Sliced Wasserstein distance from the Monte-Carlo approximated posterior
- **Computational efficiency**: Runtime in seconds

4 Simulation For Student's-t regression

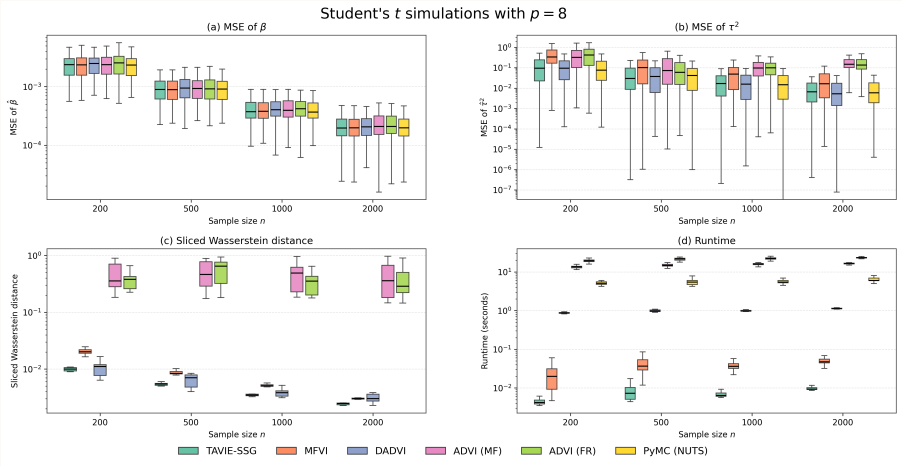


Figure 1: Student's-t regression results for fixed $p = 8$ and $n \in \{200, 500, 1000, 2000\}$

4 Simulation For Laplace regression

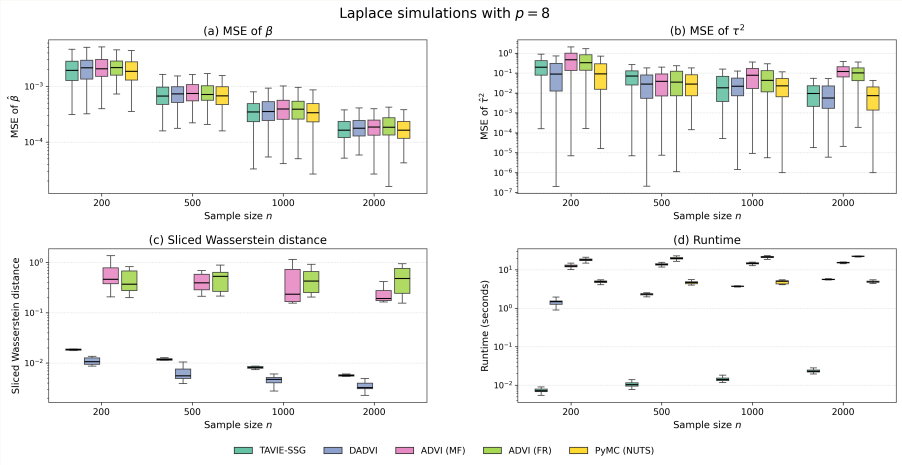


Figure 2: Laplace regression results for fixed $p = 8$ and $n \in \{200, 500, 1000, 2000\}$

4 Simulation For Negative-Binomial regression

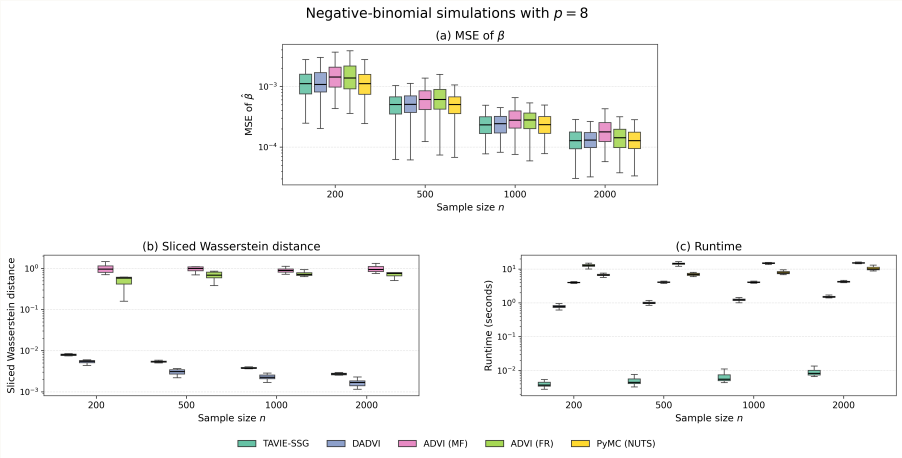
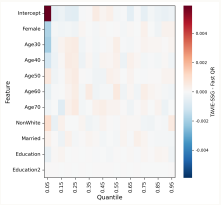


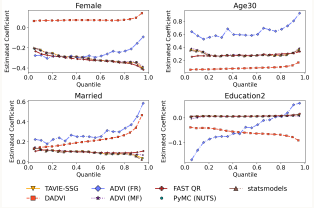
Figure 3: Negative-Binomial regression results for fixed $p = 8$ and $n \in \{200, 500, 1000, 2000\}$

4 Bayesian Quantile Regression on U.S. Census 2000 Data

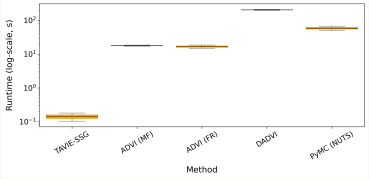
In addition to the competing methods mentioned, we ran FAST QR, a fast point estimation method for quantile on the census dataset as a baseline.



(a) Differences between TAVIE-SSG and FAST QR estimates for full data.



(b) Selected estimates from TAVIE-SSG and competitors on the subsampled ($n = 10^4$) dataset.



(c) Runtime comparison on the subsampled ($n = 10^4$) dataset.

- TAVIE-SSG Scales to the full Census dataset ($n = 5 \times 10^6$), producing estimates competitive with FAST QR across quantiles, where competing methods do not work in reasonable time.
- On subsampled data ($n = 10^4$), TAVIE-SSG estimates closely align with FAST QR and PyMC (NUTS) while being substantially faster than competing VI and MC methods.
- DADVI and full-rank ADVI exhibit substantial instability across features and quantiles.

4 Conclusions and Future Directions

- **Unified framework:** TAVIE-SSG extends tangent-based VI to broad classes of strongly super-Gaussian likelihoods.
- **Fast and trustworthy inference:** closed-form deterministic updates provide scalable posterior approximation, with convergence guarantees and near-minimax variational risk bounds.
- **Broader modeling:** extend the framework to sparsity-inducing priors, hierarchical models, and mixture models.
- **Sharper methodology:** develop tighter tangent bounds and principled model-selection procedures, and study robustness under model misspecification.

Exploit likelihood geometry to obtain scalable VI without restrictive mean-field assumptions or Monte Carlo optimization.

Thank you

Questions?

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github.com/Roy-SR-007/TAVIE-SSG

